

Question 1: Estimate the bias and variance of KNN

Random seed is 4.

a. $f(x_0) = 0.5942$;

k	1	3	5	7	9	11	13	15	17	19	21	23	25	27	29
Prediction	0.5	0.7	0.6	0.5	0.5	0.4	0.5	0.5	0.6	0.6	0.5	0.6	0.5	0.6	0.6
n	8	5	8	6	3	8	7	8	1	0	8	1	8	0	1

b.

Pattern for KNN:

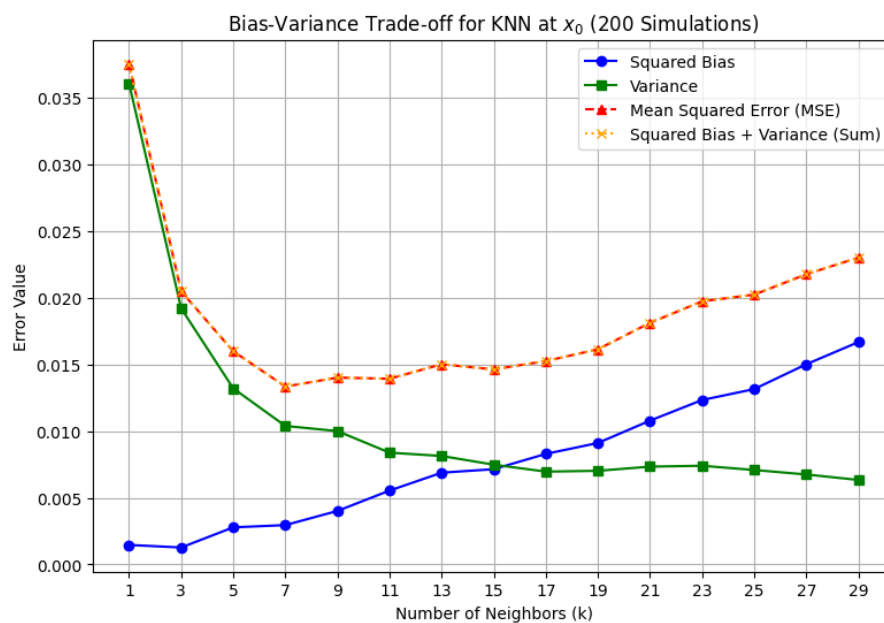
As k increases, the model's flexibility decreases.

- **Variance decreases** because averaging over a larger neighborhood (k) smooths out the sampling variability and noise of individual data points.
- **Squared Bias increases** because a larger neighborhood oversmooths the local structure, causing the prediction to drift away from the true local value $f(x_0)$ (underfitting the local neighborhood).
- The **MSE** forms a U-shape, balancing this trade-off to find an optimal k .

Comparison with Ridge / Lasso:

This trade-off is directly analogous to the tuning parameter λ in Ridge or Lasso regression.

- **Increasing λ** imposes a heavier penalty, shrinking the coefficients and making the model **simpler and more constrained**.
- Just like increasing k in KNN, **increasing λ in Ridge/Lasso increases bias and decreases variance**. Conversely, a small λ (or small k) leads to high variance and low bias.



Question 2: Compare KNN with lasso

a. Results Comparison

Setting	KNN Test MSE	Lasso Test MSE	Selected λ	Nonzero Slopes
Setting 1 (Independent)	2.126	2.065	0.1334	2
Setting 2 (Correlated, $\rho = 0.8$)	0.754	1.934	0.0422	8

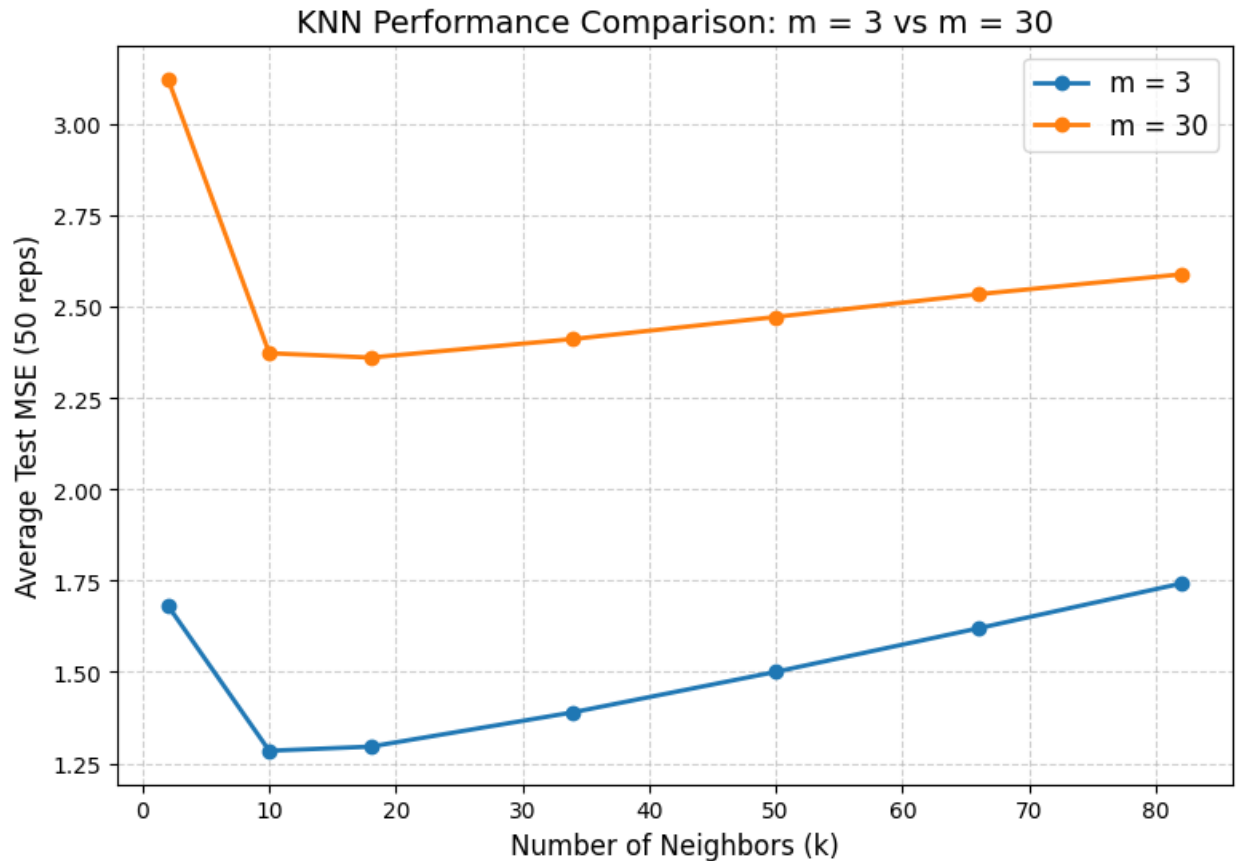
b. How Correlation Changes Euclidean Neighborhoods & Why Lasso Differs from KNN

- **Correlation & Euclidean Neighborhoods:** When features are strongly correlated (e.g., $\rho = 0.8$), data points in the feature space cluster along the direction of correlation. This compresses redundant dimensions, making true nearest neighbors denser and more clustered in Euclidean space. Consequently, KNN (a non-parametric local smoothing method) can more accurately capture local manifold structures, significantly improving prediction accuracy.
- **Why Lasso Differs:** Lasso relies on a global linear assumption and uses L_1 regularization to shrink and select coefficients (forcing irrelevant ones to zero), producing a sparse global linear combination. In contrast, KNN makes no parametric assumptions and relies entirely on local Euclidean distances to compute neighbor averages. Their underlying mechanisms are fundamentally different.

c. Impact of Non-linearity & Generalizability Claim

- **Effect of Non-linearity:** The true response model contains non-linear terms ($\sin(X_2)$ and X_3^2), whereas the Lasso fit uses only linear terms. This misspecification limits Lasso's performance, capping its MSE at the linear approximation threshold. Conversely, KNN is naturally suited for approximating non-linear functions, allowing it to heavily outperform Lasso when local structures are distinct.
- **Generalizability: No**, these results do not justify a general claim that correlation *always* helps KNN or that one method is universally better. This is because:
 1. KNN's advantage heavily depends on sample density (curse of dimensionality); under extremely high dimensions or heavy noise, high correlation can lead to redundant information and distance metric failure.
 2. If the true underlying model is linear and the feature dimension is high, Lasso's variable selection and noise-filtering capabilities typically surpass KNN. The superiority of a method depends heavily on the **form of the true underlying function (linear vs. non-linear)** and the **signal-to-noise ratio**.

Question 3: Observed dimension and latent dimension



1. Why KNN performance can still differ between $m = 3$ and $m = 30$ (even with 100 observed covariates and only two relevant latent variables)?

Although both settings share the same 100 observed dimensions and the response Y depends on only two latent variables (Z_1, Z_2), their **effective complexity and signal-to-noise ratio** differ significantly:

- When $m = 3$, the latent space is low-dimensional, meaning the variation in $\mathbf{X}$ is concentrated along fewer dominant directions.
- When $m = 30$, a much larger number of latent factors are mixed into $\mathbf{X}$ through matrix $\mathbf{A}$. This increases the intrinsic complexity and variability, making it harder for KNN to isolate the relevant signal from the irrelevant variance in high-dimensional space. Consequently, $m = 30$ generally yields a higher Test MSE.

2. What role does the noise η_i play?

- **Distance Pollution & Dilution Effect:** The noise term $\boldsymbol{\eta}_i \in \mathbb{R}^{100}$ is added independently across all 100 dimensions. In high-dimensional Euclidean distance calculations ($\|\mathbf{X}_i - \mathbf{X}_j\|^2$), the squared differences contributed by the 100 noise dimensions accumulate and dominate the distance metric.
- This blurs the true local neighborhood structure defined by the latent variables, causing the "nearest neighbors" selected by KNN to become less accurate and more noisy.

3. Why are the observed data not confined exactly to an m -dimensional linear space?

- While the deterministic component $\mathbf{AZ}_i$ lies strictly within an m -dimensional linear subspace spanned by the columns of $\mathbf{A}$, the full observation is $\mathbf{X}_i = \mathbf{AZ}_i + \boldsymbol{\eta}_i$.
- Because $\boldsymbol{\eta}_i$ is a full-rank random normal vector spanning all 100 dimensions, it perturbs the data off the m -dimensional subspace, causing the observed data $\mathbf{X}_i$ to **fill and span the entire 100-dimensional space**.

Question 4: Classify handwritten digits

1. Overall Performance and Error Rate Comparison

- **Euclidean Distance:**
 - Test errors: 61/500
 - **Test Classification Error Rate: 12.20%**
- **Manhattan Distance:**
 - Test errors: 69/500
 - **Test Classification Error Rate: 13.80%**
- **Comparison:** Under this setting and training size (first 1,000 rows), Euclidean distance performs slightly better than Manhattan distance. This indicates that squared difference penalties handle local pixel intensity variations more effectively.

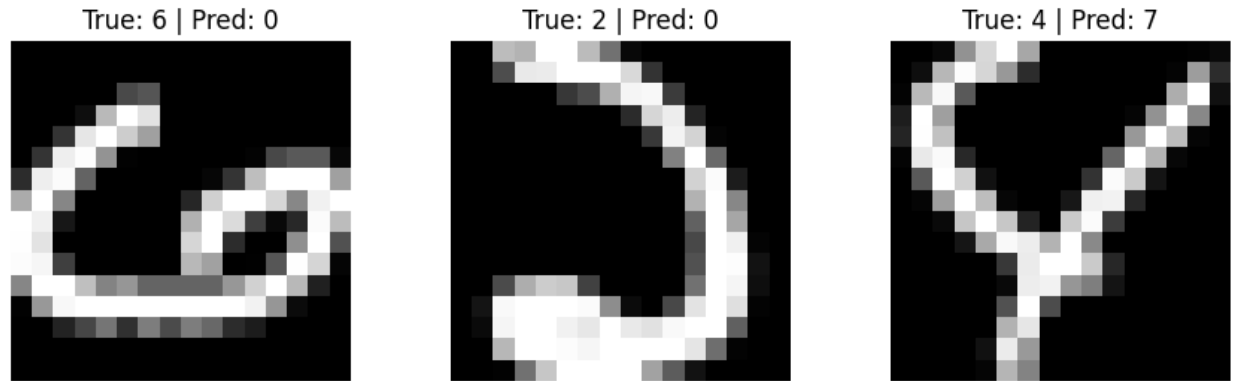
2. Confusion Matrix & Common Misclassifications

From the 10×10 confusion matrix (rows = Predicted, columns = True):

- **Most Commonly Confused Digits:**
 - **1 and 7 / 4 and 9:** Due to similar stroke structures and spatial directions, slanted or distorted digits often cross-misclassify.
 - **2 and 0 / 6 and 0:** Digits with loop structures can easily resemble the closed circle shape of 0 under irregular handwriting.

3. Visual Analysis of Euclidean Distance Errors

Euclidean 5NN - Incorrectly Classified Images



Based on the visualized incorrect test images:

1. **First Image (True: 6 | Pred: 0):**
 - **Analysis:** The top loop of the handwritten 6 is unclosed and overly rounded, forming an oval shape that resembles a 0 in the 256-dimensional pixel space.
 2. **Second Image (True: 2 | Pred: 0):**
 - **Analysis:** The bottom curve of the 2 forms an unusually bulky, closed blob, causing its pixel features to overlap more with digit 0.
 3. **Third Image (True: 4 | Pred: 7):**
 - **Analysis:** The intersection of the 4 is positioned lower down, and the stroke combination mimics the corner structure of a 7.
- **Conclusion:** These mistakes are **visually completely understandable**. KNN relies on raw pixel Euclidean distances, making it highly sensitive to global scaling, slant, stroke width, and local deformation.

4. Comparison Between Distance Measures

- **Core Conclusion:** Although both distance metrics achieve comparable overall error rates (12.20% vs 13.80%), **they make mistakes on different individual images.**
- **Reasoning:**
 - **Euclidean Distance** computes squared differences ($d_E = \sum (x_{0j} - x_{ij})^2$), which penalizes larger pixel deviations quadratically and is more sensitive to prominent local differences.
 - **Manhattan Distance** computes absolute differences ($d_M = \sum |x_{0j} - x_{ij}|$), applying linear penalties.
 - Consequently, their nearest-neighbor ranking lists differ slightly, allowing them to produce distinct prediction outcomes on specific samples despite having similar aggregate error rates.

Question 5: Create a plot-formatting skill

name: stat432-plot-formatting

description: Improve and format plots for STAT 432 homework assignments. Use this skill exclusively when the user explicitly requests plot formatting or styling assistance for STAT 432 homework.

STAT 432 Homework Plot Formatting

A guide for styling clean, professional academic data visualizations in R or Python for STAT 432 homework assignments.

When to Use

Use this skill strictly when the user explicitly requests plot formatting assistance for STAT 432 homework.

Formatting Rules (R & Python)

Sizes: Set appropriate, balanced canvas dimensions (e.g., `figsize=(8, 5)` in Matplotlib or `width=7, height=5` in R graphics) for proper visual proportions.

Margins: Ensure adequate padding and export boundaries so that all text, labels, and legends remain fully visible without clipping.

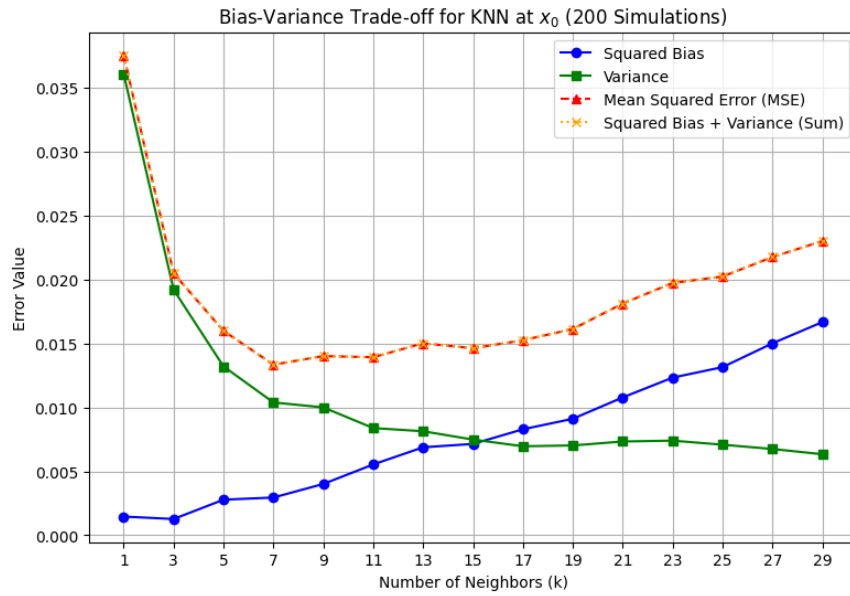
Titles: Provide concise titles formatted cleanly, placing descriptive captions below the figure in bold text with standard numbering where applicable.

Axes: Maintain minimalist and clean coordinate axes by removing distracting background grid lines and reducing or eliminating unnecessary tick marks.

Labels: Use clear, explicit variable names for axis labels and full names for legend entries, omitting redundant acronyms or suffixes.

Colors: Apply professional, low-saturation, and soft color palettes to ensure high legibility and avoid harsh visual contrasts.

Origin:



after:

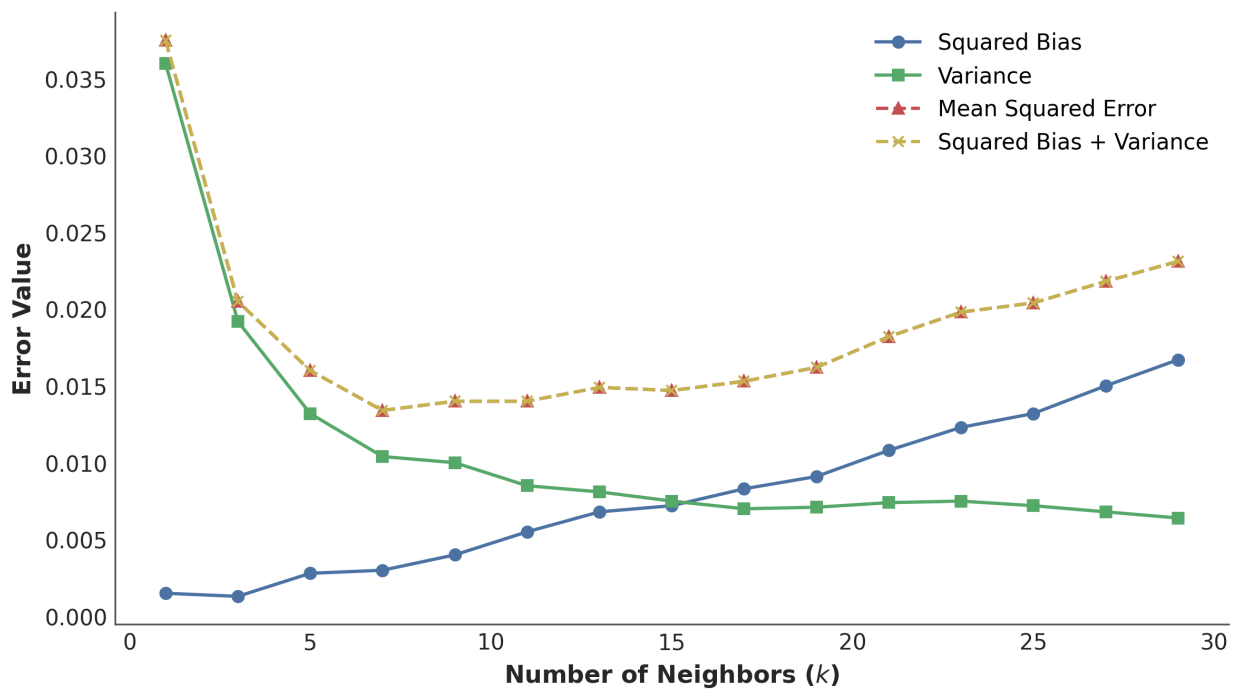


FIG. 1. Bias-variance trade-off for KNN at x_0 in 200 simulations

Design Choices and Rationale

The design choices for this plot are driven by standard academic journal guidelines to ensure a clean, professional, and publication-ready presentation:

Removal of Bounding Boxes and Grids: Most high-level journals prohibit full surrounding box lines and cluttered background grids. Removing these elements eliminates visual distractions and focuses the reviewer's attention purely on the data curves.

Figure Caption Position: Standard academic formatting requires the figure number and description (e.g., FIG. 1. ...) to be placed as a caption below the figure rather than as a title embedded inside the image area.

Low-Saturation Color Palette: High-saturation or harsh primary colors are avoided. Soft, low-saturation colors provide a polished look on digital displays and remain legible if the paper is printed in grayscale or black-and-white.

Moderate Font Sizes & 300 DPI Resolution: Font sizes are kept balanced so text remains legible without dominating the visual space when scaled for single- or double-column layouts. Furthermore, exporting the figure at 300 DPI satisfies the strict resolution requirement for high-quality printing and online publishing.